

An encyclopedia for stock markets? Wikipedia searches and stock returns

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ARTICLE INFO

Keywords:

Wikipedia
Stock market
Investor behavior
Transfer entropy

ABSTRACT

We present empirical evidence that collective investor behavior can be inferred from large-scale Wikipedia search data for individual-level stocks. Drawing upon Shannon transfer entropy, a model-free measure that considers any kind of statistical dependence between two time series, we quantify the statistical information flow between daily company-specific Wikipedia searches and stock returns for a sample of 447 stocks from 2008 to 2017. The resulting stock-wise measures on information transmission are then used as a signal within a hypothetical trading strategy. The results evidence the predictive power of Wikipedia searches and are in line with the previously documented notion of buying pressure revealed by online investor attention and the trading patterns of retail investors.

1. Introduction

A large body of finance literature is concerned with the formation of stock prices, which reflect the trading activities of both institutional and retail investors. The latter are often thought of as being at an informational disadvantage, compared to institutional investors, and are seen as noise traders in the way portrayed by Kyle (1985) or Black (1986). With widespread internet access and the rapid development of social media, online resources can increasingly be used by retail investors when searching for openly accessible company and stock market related news and information (for a recent review of the literature about social media and financial markets see, for example, Bukovina, 2016). Thus, nowadays, retail investors have far more options to gather such news and information and to build subsequent investment decisions around online sources (e.g., Chen, De, Hu, & Hwang, 2014). Moreover, consistent with the above-mentioned noise trader models and behavioral models along the lines of De Long, Shleifer, Summers, and Waldmann (1990) and Shleifer and Vishny (1997), not only has it been shown that retail investors trade in concert with other retail investors but also that return co-movements for stocks with high retail investor concentration cannot be explained by earnings forecasts and macroeconomic news (e.g., Barber, Odean, & Zhu, 2009; Kumar & Lee, 2006). As a result, data from online sources are likely to be valuable for large-scale measurements of collective (retail) investor behavior (e.g., Moat et al., 2013; Preis, Moat, & Stanley, 2013).

In the following, we specifically focus on online searches queries,

which have been shown to constitute a viable proxy for latent investor attention. On this note, Da, Engelberg, and Gao (2011) discover that the Google Search Volume Index provides a more reliable measure of investor attention than the indirect measures that have been employed in the past (for example, extreme returns and abnormal trading volume as used in Barber & Odean, 2008). Similarly, Preis et al. (2013) link changes in the search volume for financial terms on Google to broad stock market movements. Moreover, Joseph, Wintoki, and Zhang (2011) find that stock ticker searches on Google forecast abnormal stock returns and trading volume in a weekly portfolio sorting exercise and relate their results to trading activities of less sophisticated retail investors. Bijl, Kringhaug, Molnár, and Sandvik (2016) use Google search data for stock return forecasting and find that high Google search volumes lead to negative returns, contradicting the theory of investor attention generating buying pressure as suggested by Barber and Odean (2008). Turning to stock market volatility and a theoretical connection that is drawn by, for instance, Andrei and Hasler (2014), the studies of Dimpfl and Jank (2016), Hamid and Heiden (2015), and Aouadi, Aroui, and Teulon (2013) provide empirical evidence that Google searches help to improve the accuracy of realized volatility forecasts, whereas Audrino, Ballinari, and Sigrist (2019) confirm this result for a sample of individual-level stocks.

Besides Google, the online encyclopedia Wikipedia can also be used to obtain information about financial terms, events, or specific companies. While there are numerous studies relying on Google trends or searches, Wikipedia pages views or counts have rarely been used as a

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measure for investor attention. The frequent usage of Google data to measure investor attention is probably due to the popularity of Google as a search engine, it's quick and easy accessibility and the flexibility with respect to the search keywords. However, there are also caveats such as the availability of search volumes merely as an index and the potential ambiguity of search terms. These caveats might be overcome by Wikipedia data such as page views or counts (e.g. Cergol & Omladic, 2015; Kaempf, Tessenow, Kenett, & Kantelhardt, 2016; Weng, Ahmed, & Megahed, 2017).

Closely related to our paper is a study by Moat et al. (2013), who use changes in Wikipedia page views and page edits to investigate the relationship between large-scale online information gathering and subsequent stock market movements. These authors find evidence that Wikipedia searches for broad financial terms can indeed provide insights into the future development of stock market returns. However, since Moat et al. (2013) are concerned with broad search terms and stock market movements on a weekly frequency, it is not clear whether these findings hold true for company-specific Wikipedia searches on a daily frequency and a large sample of individual-level stocks.

This paper provides an investigation of the dynamic relations between company-specific Wikipedia searches and corresponding stock returns and thereby adds to previous research in three ways: (i) as the results of studies using Google searches to predict stock returns are mixed, we rely on Wikipedia data, which has only rarely been used within this context and to offer additional insights towards the relationship between online attention and stock returns; ii) we use a comprehensive data set consisting of 447 stocks over a period of 10 years to test and quantify the information transmission between online investor attention and stock returns relying on the model-free measure of transfer entropy, which is not limited to linear relationships; iii) we illustrate how our estimates can be used as a signal for trading direction within a hypothetical trading strategy.

Our results reveal that collective investor behavior can be inferred from large-scale company-specific Wikipedia search data for a group of stocks on a daily frequency. However, we find that these *Wiki-stocks* are neither characterized by extreme returns nor by abnormal trading volume, but tend to be associated with smaller market capitalization.

The remainder of the paper is structured as follows: Section 2 introduces the Shannon transfer entropy measure. Section 3 describes the data used in our empirical analysis, and results are subsequently discussed in Section 4. Next, Section 5 outlines and applies the hypothetical trading strategy that maps our estimation results to portfolio returns. Finally, Section 6 offers some concluding remarks.

2. Information flows and transfer entropy

We draw upon the concept of Shannon transfer entropy to quantify the information transfer between two time series. This measure has been heavily used in other fields such as information theory, neuroscience, or statistical physics and increasingly attracts attention of financial market researchers.¹

Let I be a discrete random variable that can take on n distinct values, where i indicates one possible outcome of I . Denoting by $p(i)$ the specific probability mass function (pmf) of I , Shannon (1948) defines a measure of uncertainty as²:

$$\eta(i) = -\log_2(p(i)), \quad (1)$$

which is large for small $p(i)$, i.e., accounting for the fact that rare outcomes convey more information. Since $0 \leq p(i) \leq 1$, it holds that $\eta(i) \geq 0$. Taking the expectation of the information content over all possible outcomes i gives the Shannon entropy measure:

$$H_I = \mathbb{E}(\eta(i)) = - \sum_i p(i) \cdot \log_2(p(i)). \quad (2)$$

Based on Shannon entropy, Schreiber (2000) proposed a measure for information transmission by measuring the reduction in uncertainty when predicting future realizations of a variable and conditioning additionally to its own past values- on the past values of a second variable. For that purpose Schreiber (2000) assumes that the dynamics of the variables over time can be described by a Markov process. He then relies on the Kullback-Leibler distance to measure the divergence of the joint probability distribution of the two series from the assumed generalized Markov property to derive a measure for information transmission, which he called transfer entropy.

Let I and J denote stationary Markov processes of order l and h , respectively, such that the probability of observing I in state i at time $t + 1$, conditional on l lags, is given by $p(i_{t+1} | i_t, \dots, i_{t-l+1}) = p(i_{t+1} | i_t, \dots, i_{t-l})$. By measuring the deviation from the generalized Markov property $p(i_{t+1} | i_t^{(l)}, j_t^{(h)}) = p(i_{t+1} | i_t^{(l)})$, where $i_t^{(l)} = (i_t, \dots, i_{t-l+1})'$ and $j_t^{(h)} = (j_t, \dots, j_{t-h+1})'$, the information transfer (or reduction in uncertainty) from process J to I is given by:

$$T_{J \rightarrow I} = \sum_i \sum_j p(i_{t+1}, i_t^{(l)}, j_t^{(h)}) \cdot \log_2 \left(\frac{p(i_{t+1} | i_t^{(l)}, j_t^{(h)})}{p(i_{t+1} | i_t^{(l)})} \right). \quad (3)$$

Thus, Shannon transfer entropy, as calculated in Eq. (3), allows to quantify the informational gain in predicting the future value of process I when observing not only past values of I but also past values of J . The information flow from I to J is quantified analogously by calculating $T_{I \rightarrow J}$, making it possible to infer the dominant direction of the information transfer between the two processes by taking the difference of $T_{J \rightarrow I}$ and $T_{I \rightarrow J}$.

As the (conditional) probabilities used to calculate transfer entropy are calculated from their empirical frequencies, the number of possible outcomes has to be limited to fit the available sample size in order to derive valid estimates. This usually implies that the time series have to be discretized before estimating Shannon transfer entropy. Following Dimpfl and Peter (2013) and the application to return time series in Behrendt, Dimpfl, Peter, and Zimmermann (2019), we subsequently partition our time series into three bins associated with extremely small, extremely large and *normal* returns in the center of the distribution. Since we are interested in general tail events, three bins should suffice for the application we have in mind. By specifying two quantiles, q_1 and q_2 , of the empirical distribution of the observed time series $\{x_t\}_{t=1}^T$ beforehand, such an encoded time series $\{S_t\}_{t=1}^T$ is obtained as follows:

$$S_t = \begin{cases} 1 & \text{for } x_t \leq q_1 \\ 2 & \text{for } q_1 < x_t < q_2 \\ 3 & \text{for } x_t \geq q_2 \end{cases} \quad \forall t. \quad (4)$$

The encoding replaces each value in $\{x_t\}_{t=1}^T$ by one of the three integers $\{1, 2, 3\}$, in line with Eq. (4). As a robustness test, these quantiles can be varied.

Empirical application of transfer entropy commonly apply a small sample correction which results in *effective* transfer entropy. This is necessary because Shannon transfer entropy estimates are constrained to positive values and consequently exhibit an upward bias which depends on the sample size. Following Marschinski and Kantz (2002), we estimate the effective transfer entropy by calculating transfer entropy with a shuffled series.³ Effective transfer entropy is then given by:

¹ See, for instance, Dimpfl and Peter (2013) and Dimpfl and Peter (2018).

² The base of the logarithm does not affect the gain in information but only the unit of measurement. By taking base 2 logarithm, the informational gain is measured in bits. In his original work, Shannon (1948) uses the natural logarithm, which amounts to information measured in nats. However, interpretation is facilitated by measurement in bits since values of $\eta(i)$ equal the optimal average number of yes/no questions one needs to ask in order to infer the outcome of I (see also, Bossomaier, Barnett, Harré, & Lizier, 2016, p. 36).

³ A shuffled time series is obtained from the encoded time series by randomly

$$ET_{J \rightarrow I} = T_{J \rightarrow I} - T_{J_{\text{shuffled}} \rightarrow I}. \quad (5)$$

$T_{J_{\text{shuffled}} \rightarrow I}$ is the Shannon transfer entropy where the time series of J is shuffled. Lastly, in order to provide statistical inference, we follow Horowitz (2003) and Dimpfl and Peter (2013) and bootstrap the underlying Markov processes I and J .

3. Data

To test for and quantify the information transfer between Wikipedia searches and individual-level stock returns, we combine data from two different sources. On the one hand, stock prices for a broad range of listed companies. On the other hand, daily count data on Wikipedia searches for each corresponding company's Wikipedia page – or, to be more precise, the number of views per Wikipedia page. Stock information is sourced from the Thomson One database and contains daily closing prices for all constituents of the Standard & Poor's 500 (S&P 500) and the Dow Jones Industrial Average (DJIA) that are part of these indices at the end of 2017.⁴ We limit our analysis to those stocks that have full data availability during the entire time period from January 2, 2008, until December 29, 2017. This leaves us with log-return time series for a total of 447 stocks with at most 2505 trading days.

Information about Wikipedia searches is sourced from the official Wikimedia data dump, which contains hourly access logs to each Wikipedia page for every language the respective page is translated into (Wikimedia Foundation, 2018c).⁵ We then aggregate the searches to daily search measures for each company. We deal with weekends by adding these searches to the each following Monday's search counts and proceeding with the residuals from regressing this series on a Monday dummy to account for the level effect. We also consider omitting weekend searches, which results in qualitatively very similar results that are available upon request. Besides, we also consider Wikipedia searches after trading hours have ended on a given trading day. The intuition behind this is as follows: We assume that retail investors might still search online for company-specific information when trading has already been concluded on a given trading day since their occupation during core working hours may not involve stock trading. In general, investors can still place orders for the next trading day after markets have closed on the current trading day. Thus, we want to make sure to capture all company-specific Wikipedia search activities on a given day. This results in 447 matching time series of daily Wikipedia searches.⁶ For each company, we label the page views series that is constructed from view counts of the English Wikipedia page *EN page views*, and the page views series containing view counts across all languages *ALL page views*. Details on the Wikipedia data are further outlined in Appendix A.

Table 1 provides average descriptive statistics for our dataset.⁷

(footnote continued)

drawing realizations from it and using these to generate a new time series. This destroys the dependencies within a bivariate context and nonzero outcomes of transfer entropy are due to finite sample bias.

⁴ In addition, daily trading volume and market capitalization are also obtained. These are used in a subsequent analysis as discussed below.

⁵ Wikimedia is both host and operator of the Wikipedia encyclopedia (Wikimedia Foundation, 2018d).

⁶ Note that there is a split in the data set due to a change in logging-behavior in 2015. From 2015 onward, Wikimedia uses different criteria to count the number of Wikipedia searches. As a result, two different types of count data are observed: Labeled as *page counts* are data for the time period from January 2, 2008, until August 5, 2016, and labeled as *page views* are data for the time period from May 1, 2015, until December 29, 2017. We combine the two series and take page views data wherever possible, resulting in a time series consisting of page counts up until April 30, 2015 and page views thereafter for each company. In order to simplify terminology, we refer to the number of views per Wikipedia page only as page views in our analysis.

⁷ Company specific descriptive statistics are available upon request.

Table 1

Descriptive statistics.

	Obs.	Mean	S.D.	Min	Max
Panel A: returns					
log-returns	1,119,722	0.000	0.023	−1.030	0.892
Panel B: absolute Wikipedia page views					
EN	1,119,722	309.602	1296.732	0.000	348,188.000
ALL	1,119,722	503.293	2012.304	0.000	408,077.000
Panel C: relative changes in Wikipedia page views					
EN	1,119,722	0.048	2.058	−784.802	783.300
ALL	1,119,722	0.052	2.123	−784.778	783.203

Generally, the series of relative changes in page views are characterized by excess kurtosis and fat tails. This supports discretizing the data as outlined in Eq. (4) and relying on tails observations to calculate transfer entropy to measure information transmission.

The table summarizes the data used in the empirical analysis. Log-returns and relative changes in Wikipedia page views are observed for 447 individual-level stocks over a time period of at most 2505 trading days (from 2008 to 01-02 until 2017-12-31) for a total of 1,119,722 observations. Shown are the number of observations, the sample mean, sample standard deviation, minimum and maximum for log-returns, the absolute Wikipedia page views (pv^a), and the relative changes in Wikipedia page views (gr^r). In every column, the respective descriptive statistic is calculated by pooling over all i and t . Panel A depicts the descriptive statistics for log-returns, Panel B for absolute Wikipedia page views, and panel C for relative changes in Wikipedia page views. We distinguish between EN and All page view series.

4. Empirical results

We consider two time periods to analyze the information transmission between Wikipedia page views and stock returns. Firstly, the time period from 2008 to 2017, comprising the whole data set, and, secondly, the time period from 2010 to 2017, excluding the financial crisis.⁸ In the following we discuss the results relying on the 5%- and 95%-quantiles to discretize the data and a Markov orders of one (i.e., y $l = h = 1$). Robustness checks using different quantiles and Markov orders have been conducted and reveal qualitatively similar results.⁹

We calculate the effective transfer entropy for information flows in both directions by shuffling 100 times and subtracting the mean over all shuffles from the Shannon transfer entropy that is calculated according to Eq. (3). Results are illustrated in Figs. 1 and 2. These graphs depict values for the estimated effective transfer entropies on the abscissa. The first row, $X \rightarrow Y$, corresponds to the effective transfer entropy estimates measuring the information transfer from relative changes in Wikipedia page views to individual-level stock returns, whereas the second row, $Y \rightarrow X$, corresponds to the estimated effective transfer entropies measuring the information transfer in the opposite direction. The last row, *Diff*, depicts the difference of the estimates from the first and second row for each stock, i.e., positive values indicating a dominant information transfer from Wikipedia searches to stock returns. The color

⁸ Code to implement Shannon transfer entropy, including the approaches to calculate effective transfer entropy measures and to conduct statistical inference, is made available through the R-package **RTransferEntropy** (for further information, see Behrendt et al., 2019).

⁹ Detailed results are available upon request. Appendix B shows results where observed time series are discretized based on 10%- and 90%-quantiles. Note that the Markov order is the same for all 447 stocks since the interpretation of results is facilitated by setting l and h to the same number of lags (Schreiber, 2000). The choice of setting l and h equal to one is guided by the finding in the literature that return and other financial time series can be adequately modeled with Markov chains of order one (e.g., McQueen & Thorley, 1991; Turner, Startz, & Nelson, 1989) and by the aim of providing a simple but robust specification.

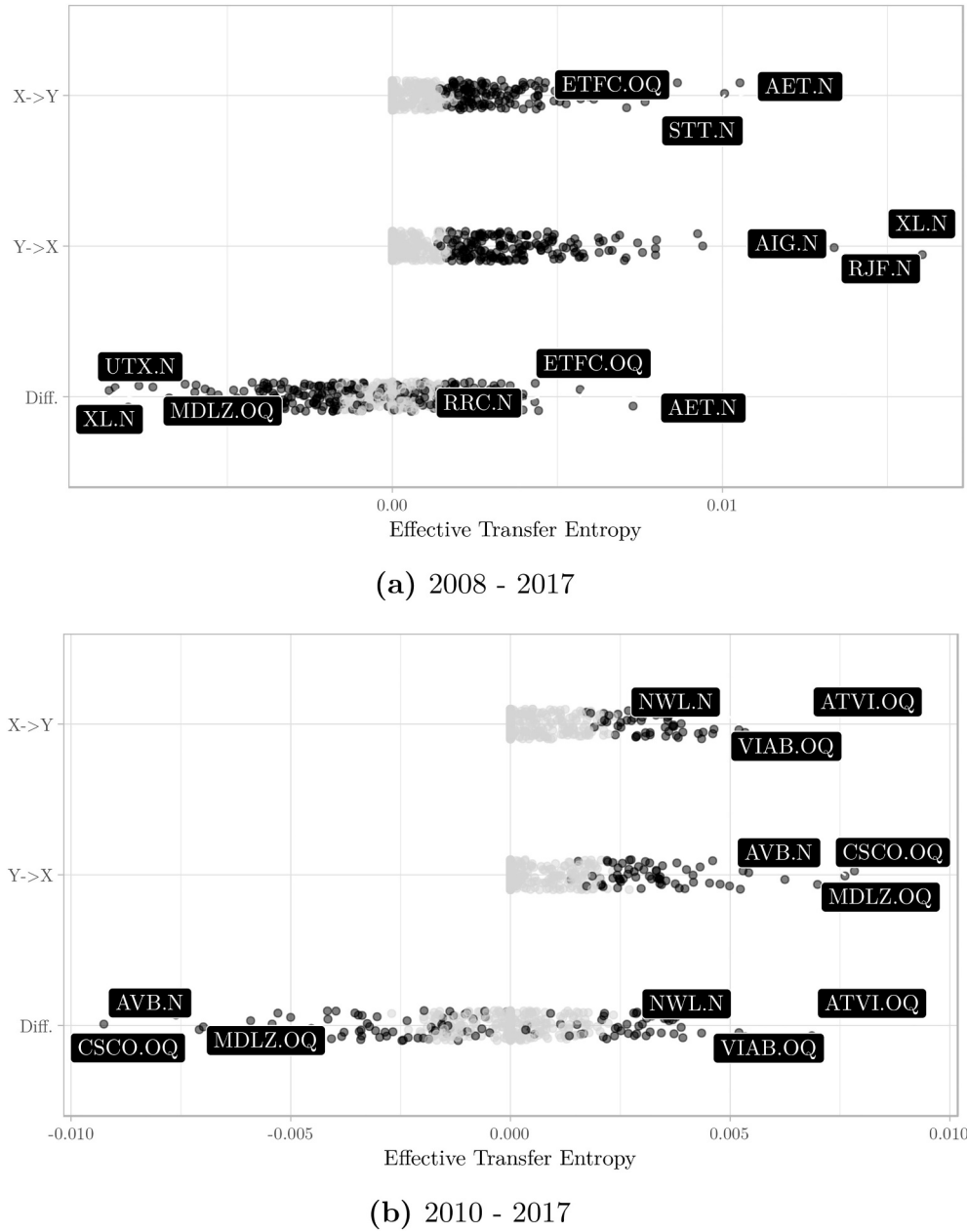


Fig. 1. Shannon transfer entropies $q(5, 95)$, gr^{ALL} series.

of the dots signals statistical significance: Black dots represent estimates that are statistically significant at the 10% significance level, grey dots represent statistically insignificant estimates. For significance in the *Diff* row, we require at least one of the estimates in $X \rightarrow Y$ or $Y \rightarrow X$ direction to be statistically significant at the 10% level. Statistical inference is based on samples of 300 bootstrapped transfer entropies for each of the 447 stocks and direction of information transfer. Fig. 1 illustrates Shannon transfer entropy estimates for the time series of relative changes in Wikipedia page views measured by gr^{ALL} and the respective return time series, (a) for the time period from 2008 to 2017 and (b) for the time period from 2010 to 2017. Fig. 2 depicts similar results for the time series of relative changes in Wikipedia page views measured by gr^{EN} .

As obvious from Figs. 1 and 2, we can group stocks into three categories: (i) stocks for which the difference of effective transfer entropies is positive and statistically significant, (ii) stocks with a statistically insignificant difference, and (iii) stocks with a negative and statistically significant difference. For the latter group we document

statistically significant information flow from returns to Wikipedia searches, which implies that extreme return movements catch the attention of investors which subsequently search for information on Wikipedia. We refer to the first group of stocks as Wiki-stocks, since for this group of stocks the dominant direction of the information transfer points from the respective Wikipedia time series to the return time series, implying that excessive online searches tend to precede extreme changes in stock prices. This is the group of stock we are interested in, which amounts to roughly 100 stocks when considering the whole period.

This figure plots the estimated effective transfer entropies for the entire sample of stocks, (a) from 2008 to 2017 and (b) from 2010 to 2017. 100 shuffles are used in the calculation. $X \rightarrow Y$ corresponds to the information transfer from Wikipedia to returns, $Y \rightarrow X$ to the information transfer from returns to Wikipedia. *Diff* depicts the difference of these estimates for each stock, positive values indicating a dominant information transfer from Wikipedia to returns. Observations are discretized based on the 5%- and 95%-quantiles and the Markov order is

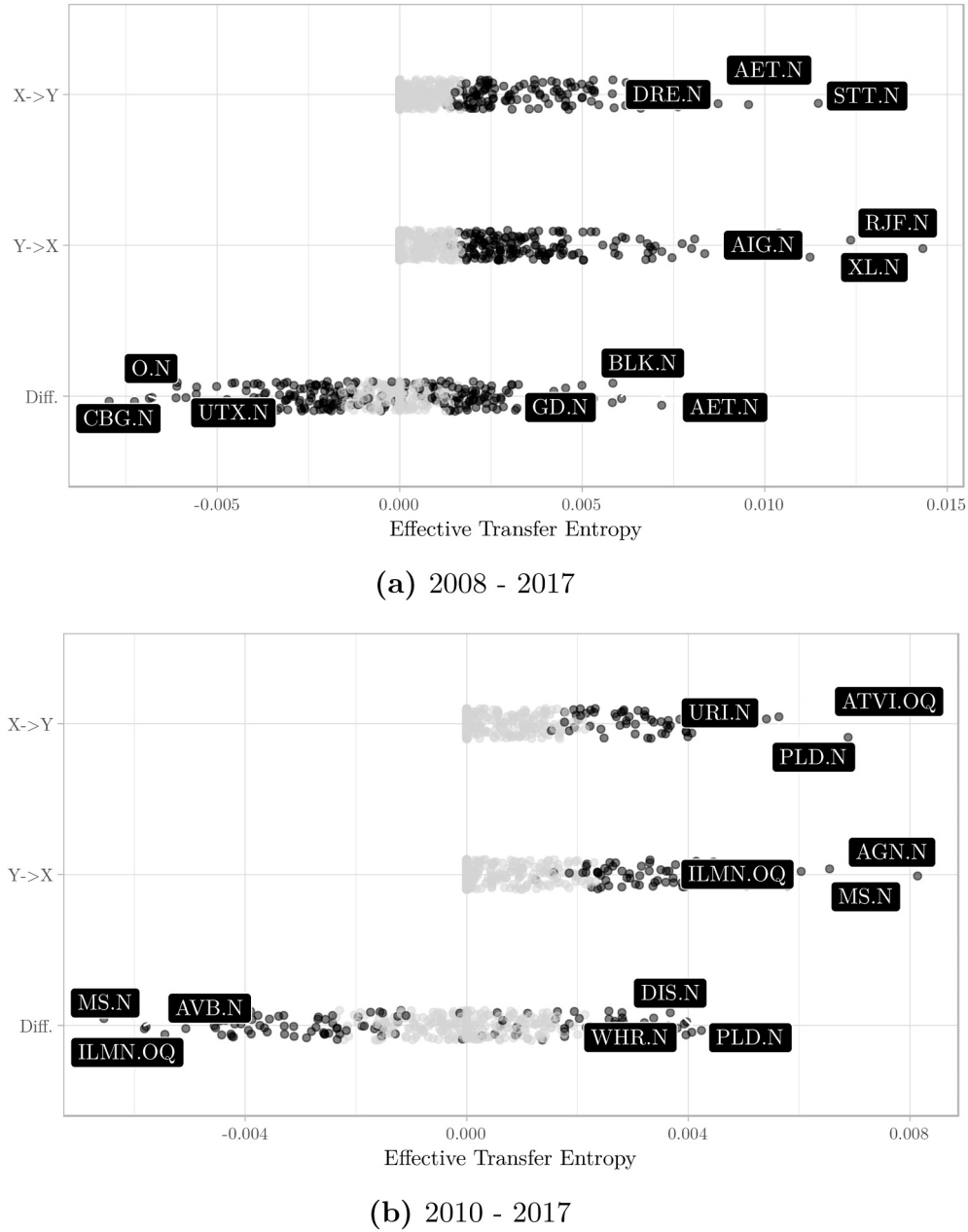


Fig. 2. Shannon transfer entropies $q(5, 95)$, gr^{EN} series.

$l = h = 1$. Black dots represent estimates that are statistically significant at the 10%-level, grey dots represent statistically insignificant estimates. Inference is based on 300 bootstrapped transfer entropies for each stock. The three largest/smallest estimates are highlighted by their corresponding ticker symbols.

This figure plots the estimated effective transfer entropies for the entire sample of stocks, (a) from 2008 to 2017 and (b) from 2010 to 2017. 100 shuffles are used in the calculation. $X \rightarrow Y$ corresponds to the information transfer from Wikipedia to returns, $Y \rightarrow X$ to the information transfer from returns to Wikipedia. *Diff* depicts the difference of these estimates for each stock, positive values indicating a dominant information transfer from Wikipedia to returns. Observations are discretized based on the 5%- and 95%-quantiles and the Markov order is $l = h = 1$. Black dots represent estimates that are statistically significant at the 10%-level, grey dots represent statistically insignificant estimates. Inference is based on 300 bootstrapped transfer entropies for each stock. The three largest/smallest estimates are highlighted by their

corresponding ticker symbols.

Consequently, for some stocks, collective investor behavior can be inferred from Wikipedia searches, similarly to the finding of Joseph et al. (2011) for company-specific Google searches. In light of this finding, it would be interesting to know what characterizes these Wiki-stocks. As mentioned above, it is likely that mostly retail investors use Wikipedia to obtain openly accessible information about companies. Retail investors have been shown to trade predominantly in stocks that grab their attention through extreme one-day returns and high abnormal trading volume (Barber & Odean, 2008) or stocks with a low market capitalization (e.g., Kumar & Lee, 2006). We therefore use three measures to classify stocks: extreme returns, abnormal trading volume, and market capitalization. Abnormal low (R_L^*) and high (R_U^*) returns are obtained by the 5%- and 95%-quantile of each stock's empirical standardized return distribution. Abnormal trading volume, AV , is calculated as in Barber and Odean (2008) by using the 5%- and 95%-quantile of the distribution of daily trading volume standardized by

Table 2
Stock characteristics, gr^{ALL} .

	+		0		−	
	Mean	St. Dev.	Mean	St. Dev.	Mean	St. Dev.
Panel A: standardization on a yearly basis						
R_L^*	−1.5549	(0.0689)	−1.5358	(0.0796)	−1.5490	(0.0741)
R_U^*	1.5833	(0.0606)	1.5627	(0.0699)	1.5759	(0.0650)
AV	1.8690	(0.1047)	1.8716	(0.1023)	1.8750	(0.0992)
Panel B: standardization on a monthly basis						
R_L^*	−1.5914	(0.0482)	−1.5937	(0.0491)	−1.5887	(0.0509)
R_U^*	1.6858	(0.0377)	1.6768	(0.0444)	1.6839	(0.0408)
AV	1.7022	(0.0932)	1.7025	(0.0849)	1.6999	(0.0819)
Panel C: market capitalization and number of estimates						
market cap	23,850		34,711		25,637	
N	92		219		136	

average daily volume of the corresponding year. Market capitalization *market cap* is given by the company's average market capitalization throughout the sample period in USD. Table 2 provide results where, for each of the three groups of stocks, the average is calculated across stocks.

On average, Wiki-stocks do not differ substantially from stocks in the other categories when referring to extreme returns and abnormal trading volume. Wiki-stocks are characterized by a lower market capitalization, in particular when considering the whole sampling period (Panel A). A potential explanation might be that small cap stocks are held by a larger share of retail investors, who might refer to Wikipedia for information, while large cap stocks commonly attract a greater share of institutional investors, who gather information from professional sources and online platforms (Boehmer & Kelley, 2009). Nevertheless, these measures do not seem to provide a holistic picture. Alternative channels are likely to link retail investor attention and Wikipedia searches. News and other (social) media sources might lead retail investors to gain first knowledge about potential companies to invest in, and they subsequently search for further information online.¹⁰

The table illustrates average values of R_L^* , R_U^* , AV, and *market cap* for three groups of stocks: (i) stocks for which the difference of effective transfer entropies is positive and statistically significant (+), (ii) stocks with a statistically insignificant difference (0), and (iii) stocks with a negative and statistically significant difference (−). Additionally, standard deviations are given in parentheses for R_L^* , R_U^* , and AV. Shannon transfer entropy estimates are taken from Fig. 1(a). Thus, relative changes in Wikipedia page views are measured by gr^{ALL} for the time period from 2008 to 2017 (Panel A) and 2010–2017 (Panel B).

5. A simple application to stock trading

We design a hypothetical trading strategy for Wikipedia stocks that uses retail investor attention as measured by a changes in page views as a trading signal. This approach is in line with Preis et al. (2013) and Moat et al. (2013), who utilize investment strategies to infer collective investor behavior from online search data. See also the portfolio sorting exercise based on stock ticker searches on Google conducted by Joseph et al. (2011). Thus, the goal is not to design a realistic and profitable trading strategy but to use a simple framework to test whether the information transmission from Wikipedia page views to stock returns implied by significant transfer entropy estimates documented in the previous section can be translated into buying and selling signals. Trading on these signals should consequently yield larger returns than

randomly buying and selling the same sample of Wikipedia stocks.

In detail, the steps are as follows: Firstly, Wiki-stocks are selected from the sample of 447 stocks, as identified by the transfer entropy estimates in the previous section. As a robustness check, transfer entropy estimates corresponding to a more conservative significance level of 5% are also considered. Secondly, k unique Wiki-portfolios of size m are constructed. For the stocks in these k portfolios, it is recorded whether or not the relative change in Wikipedia page views (gr_{it}^{EN} and gr_{it}^{ALL}) for company i , $1 \leq i \leq m$, on trading day t is greater than or equal to zero. This is considered as an increase in investor attention and a buying signal, so the stock is bought at the closing price p_t on trading day t . The position is closed by selling the stock at the closing price p_{t+1} on the following trading day. If the relative change in Wikipedia page views is less than zero, this is considered as a selling signal and the stock is sold at the closing price p_t on trading day t and subsequently bought at the closing price p_{t+1} on the following trading day. Transaction costs are ignored for simplicity (in line with the applications in Joseph et al., 2011; Preis et al., 2013; Moat et al., 2013), and one opening and one closing transaction per trading day are allowed. Moreover, the possibility to sell short is taken into account, i.e., the possibility to sell a stock without owning it. Thus, one share of each stock can be bought and sold on a given trading day. Thirdly, stock returns are calculated as $\ln(p_{t+1}) - \ln(p_t)$ if a long position is taken and as $\ln(p_t) - \ln(p_{t+1})$ if a short position is taken. Daily portfolio returns are cumulated over the entire trading period. Lastly, average cumulative portfolio returns are obtained by averaging returns over all k portfolios that are traded.

These returns are compared with the returns that are realized by the same k portfolios of size m in the case that trading is conducted randomly. This implies that each trade is randomly selected to be a buy or sell, closing the position accordingly on the next day. In the following, these portfolios are referred to as random-portfolios in contrast to the Wikipedia-portfolios outlined above. Please note that both portfolio categories are based on the same stocks, i.e., there is no bias due to stock specific risk factors.

Since the effect of Wikipedia search momentum likely takes time to manifest itself in the form of average cumulative portfolio returns, i.e., it takes some time for retail investors to place orders after having obtained company-specific information online, a variation of the above trading strategy is entertained where a transaction occurs only on the first trading day of a given week. Thus, a weekly trading pattern is introduced. Denote by τ the time grid of observations corresponding to the first trading day of a week (assuming this to be a Monday in every week for notational convenience), i.e., $\tau = 1, 6, 11, \dots$, then stock returns are calculated as $\ln(p_{\tau+5}) - \ln(p_\tau)$ if a long position is taken and as $\ln(p_\tau) - \ln(p_{\tau+5})$ if a short position is taken. In order to use all available information, the type of transaction is now determined by the average of the daily relative changes in Wikipedia page views during the previous trading week. If the average relative change in Wikipedia page views is greater than or equal to zero, a long position is taken and a short position otherwise.

Table 3 summarizes the results of this trading strategy. Average cumulative portfolio returns, given in basis points, are calculated for the time period from 2008 to 2017 over $k = 100$ portfolios across different portfolio sizes m , where $m \in \{10, 20, 30, 40, 50\}$, as specified in the first column. The Wikipedia page views series used in the transfer entropy estimation are denoted by (ALL) and (EN), i.e., gr^{ALL} and gr^{EN} . Columns two to five refer to portfolios constructed according to the results illustrated in Fig. 1(a) and a daily as well as a weekly trading pattern. As stated above, not only the transfer entropy estimates in Fig. 1(a) are considered to construct portfolios (Panel A) but also transfer entropy estimates that correspond to a significance level of 5% (Panel B). Columns six to nine depict the equivalent results for portfolios constructed from transfer entropy estimates taken from Fig. 2(a). The same exercise is repeated for the post-crisis period from 2010 until 2017, and results can be found in Table 4. Due to a lower number of

¹⁰ Regarding the earnings announcements of companies, Drake, Roulstone, and Thornock (2012) show that investors' online information demand prior to these announcements, as measured in terms of Google searches, is positively associated with media attention and news.

Table 3

Average cumulative portfolio returns of Wiki- and random-portfolios in hypothetical trading strategy (2008–2017).

Size	(ALL) daily		(ALL) weekly		(EN) daily		(EN) weekly	
	Wiki	Random	Wiki	Random	Wiki	Random	Wiki	Random
Panel A: transfer entropies at a significance level of 10%								
10	2.107	0.165	8.695	0.898	2.152	0.161	8.471	−0.862
20	1.818	−0.010	8.588	0.776	2.040	−0.120	8.462	0.120
30	1.912	0.008	8.426	0.308	1.962	−0.115	8.206	0.314
40	1.897	−0.097	7.921	0.726	1.929	−0.029	8.778	0.248
50	1.888	−0.086	8.099	0.646	1.972	0.022	8.247	−0.122
Panel B: transfer entropies at a significance level of 5%								
10	2.069	0.009	8.395	−1.484	2.247	0.320	8.278	1.339
20	2.066	0.030	8.419	0.861	2.265	0.200	9.666	−0.754
30	1.975	0.113	8.256	0.460	2.310	0.041	9.669	0.814
40	1.921	0.060	8.198	0.052	2.355	−0.031	9.839	0.350
50	2.015	−0.002	8.423	−0.369	2.300	0.019	9.641	−0.146

Table 4

Average cumulative portfolio returns of Wiki- and random-portfolios in hypothetical trading strategy (2010–2017).

Size	(ALL) daily		(ALL) weekly		(EN) daily		(EN) weekly	
	Wiki	Random	Wiki	Random	Wiki	Random	Wiki	Random
Panel A: transfer entropies at a significance level of 10%								
5	1.931	0.044	16.187	0.413	1.400	0.051	15.244	−0.769
10	1.797	−0.034	16.412	0.268	1.451	−0.014	15.755	0.756
15	1.842	0.032	17.445	−0.045	1.377	0.055	15.769	0.054
20	1.775	−0.076	17.148	0.792	1.371	−0.083	15.136	−0.242
25	1.914	−0.046	17.686	−0.581	1.458	0.044	15.537	0.321
Panel B: transfer entropies at a significance level of 5%								
5	1.873	−0.142	19.014	0.607	1.938	0.242	17.537	3.771
10	1.891	0.071	19.182	1.221	1.628	0.027	16.460	0.227
15	1.947	−0.054	19.408	−0.822	1.662	0.025	16.212	0.415
20	1.923	−0.087	19.502	0.137	1.717	−0.209	16.459	0.209
25	1.948	−0.024	19.409	−0.314	1.677	0.002	16.333	−0.294

Wiki-stocks for this time period, the portfolio size is restricted such that $m \in \{5, 10, 15, 20, 25\}$. Tables C.1 to C.6 in the Appendix show the standard deviation, skewness and kurtosis of portfolio returns. Results on portfolios based on different quantiles when discretizing the data to derive transfer entropy estimates are available upon request.

Results are consistent across all portfolio sizes and trading patterns: Average cumulative portfolio returns of Wiki-portfolios are larger than returns from random-portfolios. This finding is even more pronounced if trading is conducted on a weekly basis. In the case of a weekly trading pattern, average cumulative portfolio returns are even larger for the post-crisis period. This result still holds when accounting for the higher standard deviation associated with the Wiki-portfolios as visible in Tables C.1 and C.2. Overall, trading on Wikipedia searches generates moderate gains on average and a more extended holding period increases the returns from Wikipedia search momentum. The assumption that retail investors first obtain some openly accessible information about the companies they are going to invest in and that they need more time to place orders than institutional investors can be replicated in the trading strategy. Thus, results linked to Wikipedia search momentum are in line with the trading patterns of retail investors. We acknowledge that a more careful (i.e., non-random) collection of stocks is likely to improve performance considerably, however, designing a superior trading strategy is not the purpose of this exercise. Overall, our simple trading strategy helps to illustrate the direction of the effect that

is implied by the transfer entropy estimates from the previous section.

The table summarizes average cumulative portfolio returns (in basis points) of $k = 100$ Wiki- and random-portfolios with varying portfolio size m . Wiki- and random-portfolios are constructed based on the results of the Shannon transfer entropy estimation in Figs. 1(a) and 2(a), here denoted by (ALL) and (EN), respectively. Both a daily and weekly trading pattern is considered. The entire time span from 2008 to 2017 is used to calculate overall cumulative portfolio returns, which are then averaged across all k portfolios. Panel A refers to results of portfolios constructed based on the transfer entropy estimates depicted in Figs. 1(b) and 2(b). Panel B applies a more conservative significance level of 5% to the transfer entropy estimates on which portfolio construction is subsequently based.

The table summarizes average cumulative portfolio returns (in basis points) of $k = 100$ Wiki- and random-portfolios with varying portfolio size m . Wiki- and random-portfolios are constructed based on the results of the Shannon transfer entropy estimation in Figs. 1(a) and 2(a), here denoted by (ALL) and (EN), respectively. Both a daily and weekly trading pattern is considered. The time span from 2010 to 2017 is used to calculate overall cumulative portfolio returns, which are then averaged across all k portfolios. Panel A refers to results of portfolios constructed based on the transfer entropy estimates depicted in Figs. 1(b) and 2(b). Panel B applies a more conservative significance level of 5% to the transfer entropy estimates on which portfolio construction is subsequently based.

6. Conclusion

In this paper, we present empirical evidence that for a significant share of our sample stocks, collective investor behavior can be inferred from large-scale Wikipedia search data for individual-level stocks. Using Shannon transfer entropy, a model-free measure that considers any kind of statistical dependence between the time series of relative changes in Wikipedia page views and stock returns, we can infer on the dominant direction of the information transfer for each company's stock. It is possible to identify Wiki-stocks, i.e., stocks for which the dominant direction of the information transfer points from Wikipedia searches to returns. While Wiki-stocks have a lower market capitalization on average, comparing them to the remaining stocks in our sample based on extreme returns and abnormal trading volume reveals that Wiki-stocks do not differ substantially from these remaining stocks. Thus, there have to be other channels through which Wiki-stocks attract the attention of potential investors. In order to express our findings in terms of monetary values, we design a hypothetical trading strategy based on Wikipedia search momentum that maps our findings to average portfolio returns and provides some economic interpretation for the transfer entropy estimates. We show that it takes time for the momentum effect of Wikipedia searches to manifest itself in the form of higher returns. This is in line with the overall notion of retail investors' trading patterns.

Given the increasing data availability, future research should further refine the link between online search data and data from social media platforms on the one hand and retail investor behavior on the other hand. This would broaden our understanding of the feedback loops between online sources and financial market outcomes. Most importantly, this relates to the question of how to classify stocks for which the statistical information transfer from online searches to stock returns dominates and, ultimately, to the question what constitutes the driving force behind this online search momentum.

Acknowledgements

Foundation (Grant PE 2370/2-1). The views expressed in this paper do not necessarily represent the views of d-fine GmbH.

Simon Behrendt acknowledges funding by the German Research

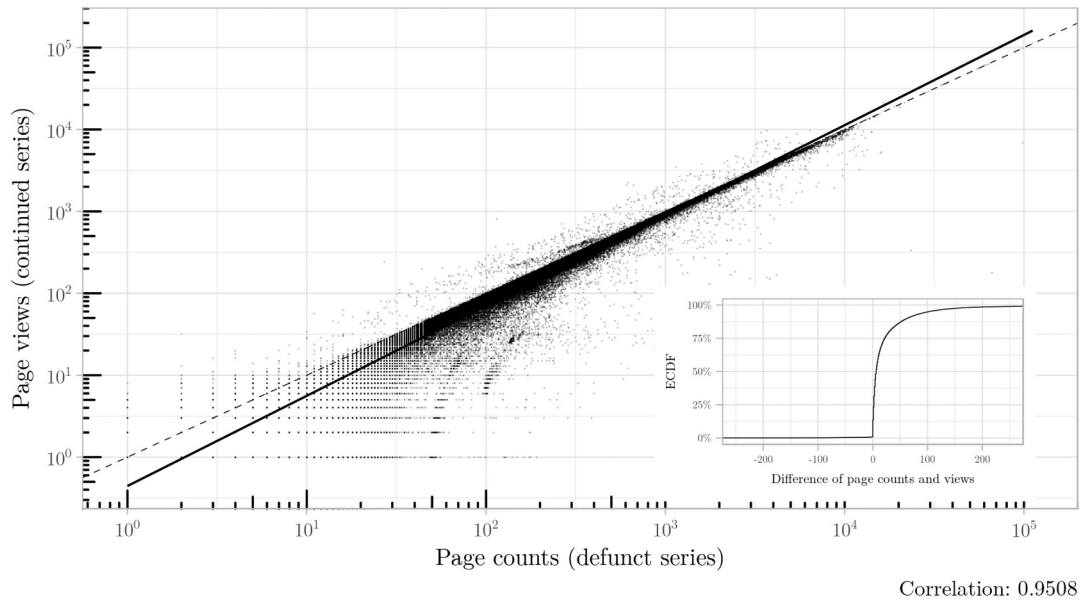
Appendix A. Details on Wikipedia data

For each of the 447 companies, we identify the page-key that is associated with the respective company's English Wikipedia page, which might include multiple pages per component (e.g., given different spelling, treatment of special characters, or the “redirected from”-section). For example, the Wikipedia page for the company American Express includes page view data for the keys “American_Express” and “American_Express.Co”, which are both redirected to the former key. Some other companies have special characters in their keys, leading to two different key versions. For example, Deere & Company, for which the keys are “Deere_%26_Co.” and “Deere_&Co.”. Additionally, if a Wikipedia page is translated into languages other than English and these pages use the identical page-key, we gather that data into a second page views time series containing all views for that key. For example, the company 3 M has the key “3 M” in all languages.¹¹ Accordingly, for each company, we label the page views series that is constructed from view counts of the English Wikipedia page *EN page views*, and the page views series containing view counts across all languages *ALL page views*.¹²

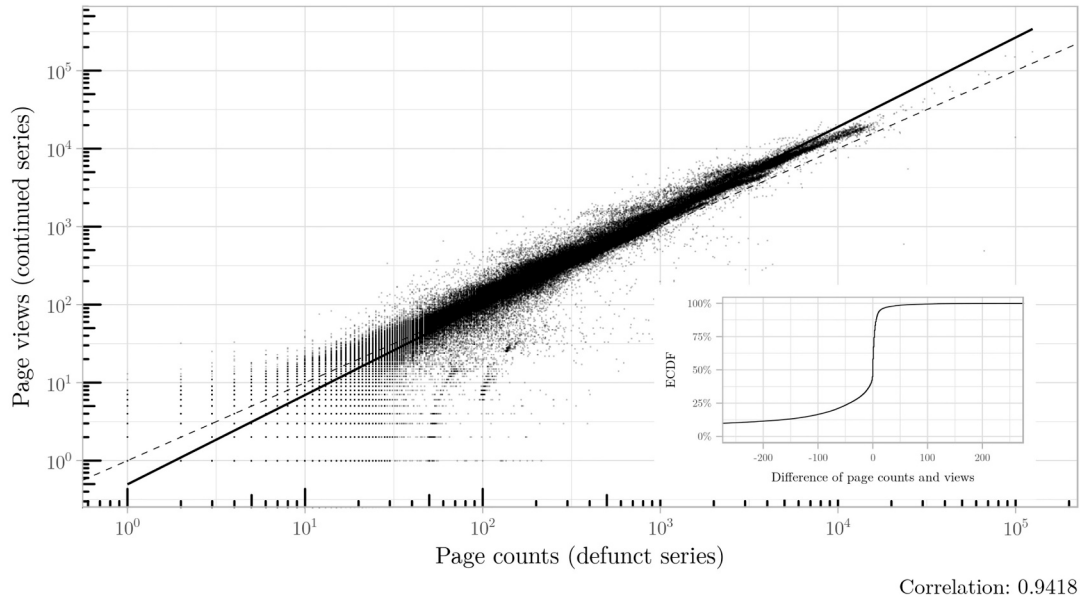
In our raw Wikipedia data, we observe two different types of count data, namely page counts and page views. While we take page views data wherever possible, we also compare all observations of the two time series for each company in our sample (EN page counts and page views as well as ALL page counts and page views) in the overlapping window comprising 463 trading days and 206,961 observations per series across all 447 stocks. Fig. A.1 illustrates the results of this comparison in the form of two scatter plots for (a) EN page counts and views and (b) ALL page counts and views across all companies. Most observations lie below the dashed line, indicating that the shift from page counts to page views reduces the logged amount of Wikipedia traffic. This is more pronounced for EN page counts and views than for ALL page counts and views and in accordance with the official description, which states that the switch in the view count variables should remove traffic originating from bots (for a detailed definition of a page count and a page view, see [Wikimedia Foundation, 2018a, 2018b](#)). Overall, these plots highlight the similarity of both count variables for all observations. The series are highly correlated with a coefficient of 95% and 94.2%, respectively. Moreover, both plots contain an additional empirical cumulative distribution function (ECDF) of the difference between page counts and page views. The ECDFs further illustrate that only about 1% of all observations have more page views than page counts in the EN page counts and views series and around 46% in the ALL page counts and views series. Most observations have a difference between page counts and page views series of less than 20 per day with 72.6% in the EN page counts and views series and 65.3% when considering the ALL page counts and views series. With these results at hand, the approach to take page views data wherever possible seems to be justified and is unlikely to introduce a significant bias in the empirical analysis.

¹¹ With respect to the time frame of our analysis, the languages in which 3 M has a Wikipedia page include Danish, German, English, Spanish, French, Italian, Dutch, Polish, Russian, and Turkish.

¹² The entire data set for the ten years consists of roughly 10 TB. In order to handle this amount of data, we filter for the respective company pages using bash-scripts and the GNU parallel library (for more information, see [Tange, 2011](#)).



(a) EN page counts and views



(b) ALL page counts and views

Fig. A.1. Comparison of page views and page counts.

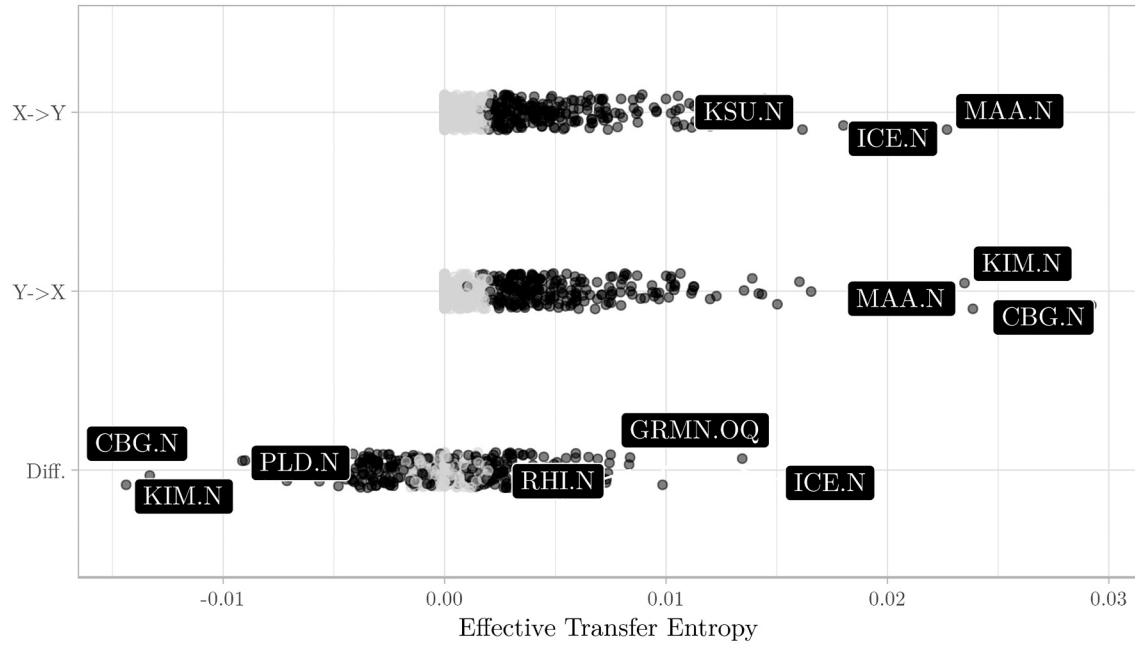
Both plots depict a scatter plot of page counts and page views in the overlapping time window from 2015 to 05-01 until 2016-08-05, for (a) EN page counts and views and (b) ALL page counts and views. The axes are log-scaled, and dashed lines in both plots indicate a correlation of 100%. The solid lines represent the best fit at a correlation of 95% (EN) and 94.2% (ALL), respectively. Moreover, both plots contain an additional empirical cumulative distribution function (ECDF) of the difference between page counts and page views.

To estimate Shannon transfer entropy, page views are converted from count to continuous data that is stationary for each time series. Stationarity tests for continuous variables are straightforward to implement, and there is no loss in information by transforming page views. Denote by gr_{it}^s the relative change in Wikipedia page views for company i , $1 \leq i \leq 447$, from trading day $t - 1$ to trading day t , where s either refers to the EN series or the ALL series. Page views for company i on day t are given by pv_{it}^s . The change in page views is scaled by the sample mean of the page views series of company i , denoted by $\bar{pv}_i^s$. The measure by which the data is scaled does not matter since estimation involves discretizing the time series into three bins. However, the sample mean is positive for every company, which means that no observations are lost by scaling with the sample mean. Relative changes in Wikipedia page views for each company and both EN and ALL page view series are obtained as

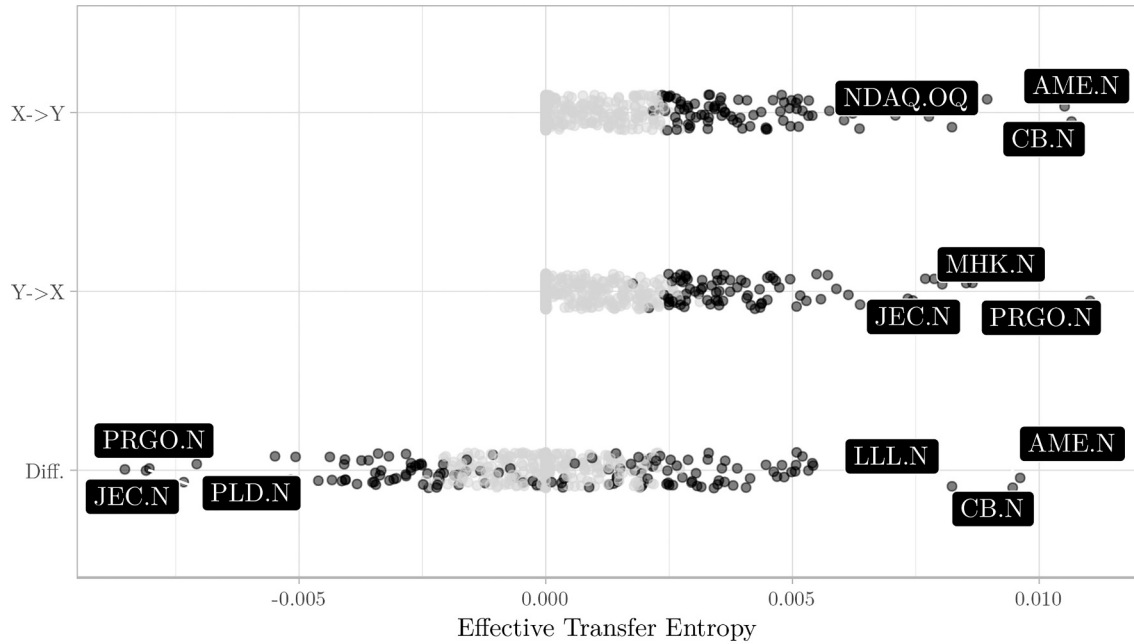
$$gr_{it}^s = \frac{pv_{it}^s - pv_{it-1}^s}{\bar{pv}_i^s}. \quad (6)$$

Calculating usual growth rates or log-changes is not feasible in the given case since some companies have zero page views for some t . All gr_{it}^s series are stationary for every company i , according to augmented Dickey-Fuller tests.

Appendix B. Transfer entropy plots



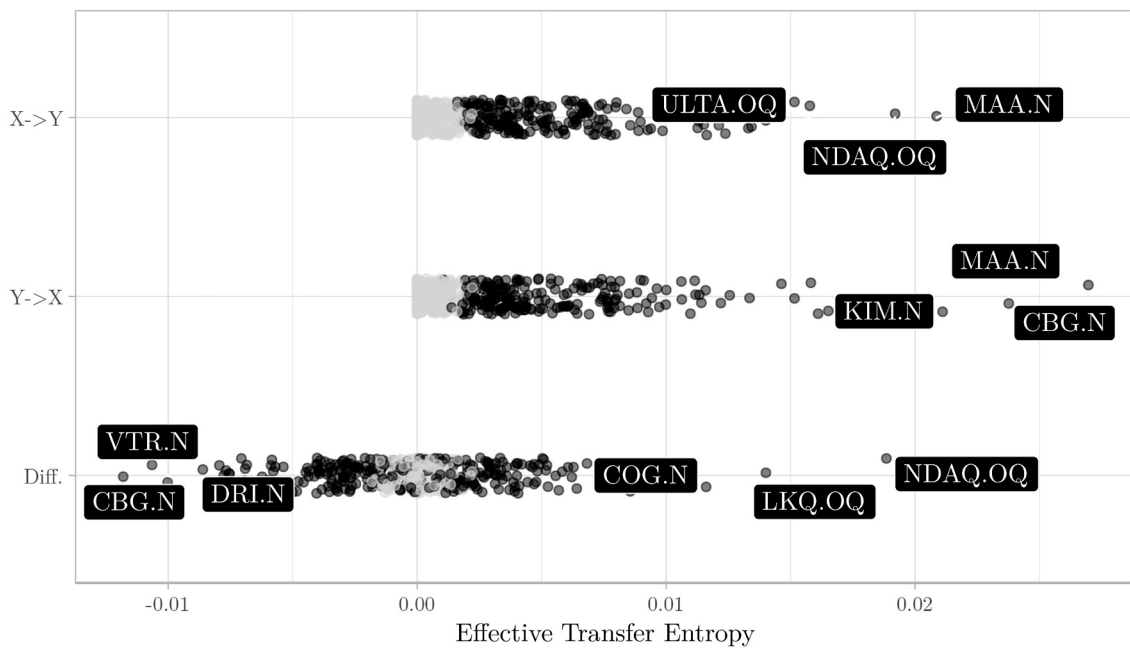
(a) 2008 - 2017



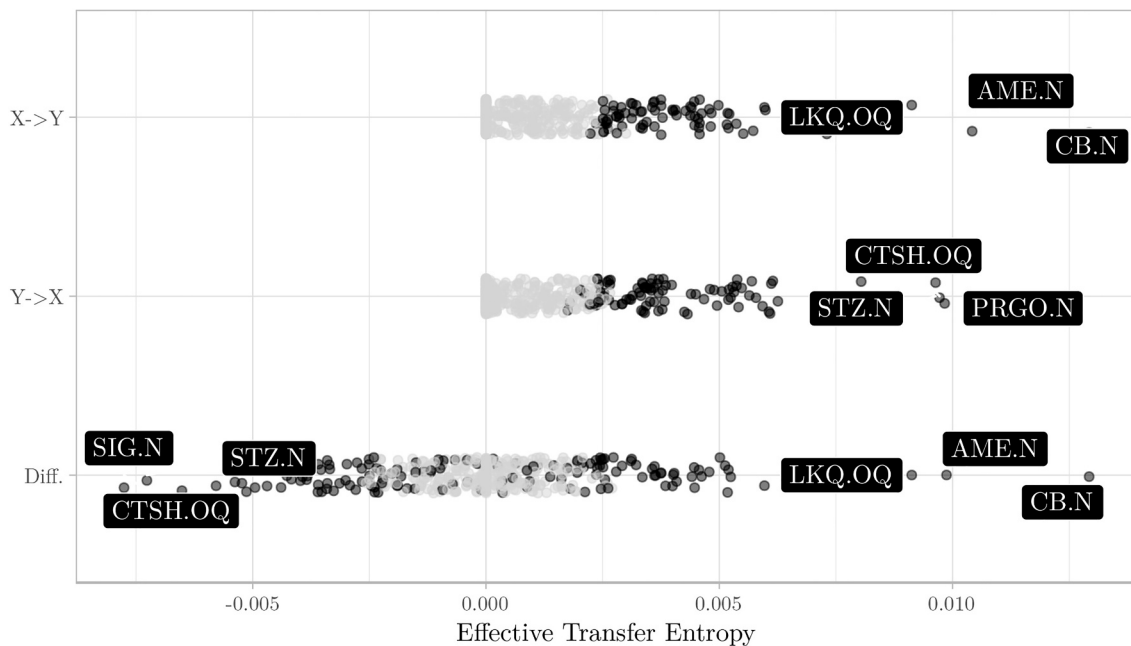
(b) 2010 - 2017

Fig. B.2. Shannon transfer entropies $q(10, 90)$, g^{ALL} series.

This figure plots the estimated effective transfer entropies for the entire sample of stocks, (a) from 2008 to 2017 and (b) from 2010 to 2017. 100 shuffles are used in the calculation. $X \rightarrow Y$ corresponds to the information transfer from Wikipedia to returns, $Y \rightarrow X$ to the information transfer from returns to Wikipedia. Diff. depicts the difference of these estimates for each stock, positive values indicating a dominant information transfer from Wikipedia to returns. Observations are discretized based on the 10%- and 90%-quantiles and the Markov order is $l = h = 1$. Black dots represent estimates that are statistically significant at the 10%-level, grey dots represent statistically insignificant estimates. Inference is based on 300 bootstrapped transfer entropies for each stock. The three largest/smallest estimates are highlighted by their corresponding ticker symbols.



(a) 2008 - 2017



(b) 2010 - 2017

Fig. B.3. Shannon transfer entropies $q(10, 90)$, gr^{EN} series.

This figure plots the estimated effective transfer entropies for the entire sample of stocks, (a) from 2008 to 2017 and (b) from 2010 to 2017. 100 shuffles are used in the calculation. $X \rightarrow Y$ corresponds to the information transfer from Wikipedia to returns, $Y \rightarrow X$ to the information transfer from returns to Wikipedia. *Diff* depicts the difference of these estimates for each stock, positive values indicating a dominant information transfer from Wikipedia to returns. Observations are discretized based on the 10%- and 90%-quantiles and the Markov order is $l = h = 1$. Black dots represent estimates that are statistically significant at the 10%-level, grey dots represent statistically insignificant estimates. Inference is based on 300 bootstrapped transfer entropies for each stock. The three largest/smallest estimates are highlighted by their corresponding ticker symbols.

Appendix C. Trading strategy

Table C.1

Standard deviation of portfolio returns of Wiki- and random-portfolios in hypothetical trading strategy (2008–2017).

Size	(ALL) daily		(ALL) weekly		(EN) daily		(EN) weekly	
	Wiki	Random	Wiki	Random	Wiki	Random	Wiki	Random
Panel A: transfer entropies at a significance level of 10%								
10	105.729	81.685	289.378	184.548	103.243	80.034	279.532	179.790
20	90.051	57.407	263.739	130.945	87.507	56.789	251.336	127.113
30	83.234	46.571	249.653	105.892	81.399	46.231	241.476	102.956
40	79.843	40.482	246.202	91.904	78.404	39.964	238.415	88.936
50	78.094	35.860	243.257	81.631	76.821	35.465	236.650	79.000
Panel B: transfer entropies at a significance level of 5%								
10	107.079	82.208	298.173	187.039	108.748	83.602	305.948	187.861
20	91.090	57.659	271.227	130.703	92.942	58.980	280.032	131.961
30	85.453	47.132	261.711	107.606	86.531	49.105	271.690	109.186
40	82.521	41.088	256.639	91.875	83.252	42.198	266.534	94.521
50	80.923	36.547	254.918	84.235	81.100	37.848	263.216	85.524

The table summarizes average cumulative portfolio returns (in basis points) of $k = 100$ Wiki- and random-portfolios with varying portfolio size m . Wiki- and random-portfolios are constructed based on the results of the Shannon transfer entropy estimation in Figs. 1(a) and 2(a), here denoted by (ALL) and (EN), respectively. Both a daily and weekly trading pattern is considered. The entire time span from 2008 to 2017 is used to calculate overall cumulative portfolio returns, which are then averaged across all k portfolios. Panel A refers to results of portfolios constructed based on the transfer entropy estimates depicted in Figs. 1(a) and 2(a). Panel B applies a more conservative significance level of 5% to the transfer entropy estimates on which portfolio construction is subsequently based.

Table C.2

Standard deviation of portfolio returns of Wiki- and random-portfolios in hypothetical trading strategy (2010–2017).

Size	(ALL) daily		(ALL) weekly		(EN) daily		(EN) weekly	
	Wiki	Random	Wiki	Random	Wiki	Random	Wiki	Random
Panel A: transfer entropies at a significance level of 10%								
5	87.848	79.018	232.557	188.017	97.932	88.208	261.366	212.177
10	68.546	55.142	193.501	131.508	76.530	62.364	217.511	146.745
15	61.093	45.428	182.372	107.277	68.598	50.765	205.413	121.554
20	56.841	39.220	173.924	92.796	63.346	43.696	195.340	104.187
25	54.170	35.026	169.505	83.226	60.258	38.879	189.755	93.033
Panel B: transfer entropies at a significance level of 5%								
5	90.039	81.222	242.054	191.264	93.849	85.350	249.582	203.324
10	71.699	57.925	210.189	135.707	75.447	60.756	220.481	145.559
15	63.945	47.538	195.653	113.363	66.645	49.514	201.479	118.379
20	60.278	41.418	191.441	97.821	62.621	42.897	196.661	102.165
25	57.453	36.926	186.353	87.284	60.097	39.322	191.527	93.748

The table summarizes average cumulative portfolio returns (in basis points) of $k = 100$ Wiki- and random-portfolios with varying portfolio size m . Wiki- and random-portfolios are constructed based on the results of the Shannon transfer entropy estimation in Figs. 1(b) and 2(b), here denoted by (ALL) and (EN), respectively. Both a daily and weekly trading pattern is considered. The time span from 2010 to 2017 is used to calculate overall cumulative portfolio returns, which are then averaged across all k portfolios. Panel A refers to results of portfolios constructed based on the transfer entropy estimates depicted in Figs. 1(b) and 2(b). Panel B applies a more conservative significance level of 5% to the transfer entropy estimates on which portfolio construction is subsequently based.

Table C.3

Skewness of portfolio returns of Wiki- and random-portfolios in hypothetical trading strategy (2008–2017).

Size	(ALL) daily		(ALL) weekly		(EN) daily		(EN) weekly	
	Wiki	Random	Wiki	Random	Wiki	Random	Wiki	Random
Panel A: transfer entropies at a significance level of 10%								
10	0.322	0.149	−0.565	0.072	0.456	0.004	−0.396	−0.145
20	0.528	−0.039	−0.536	0.028	0.818	−0.082	−0.310	−0.058
30	0.674	0.018	−0.475	−0.141	0.909	0.042	−0.348	−0.143
40	0.759	−0.059	−0.477	−0.086	1.063	−0.035	−0.256	−0.026
50	0.862	−0.088	−0.409	0.031	1.077	−0.029	−0.255	0.061
Panel B: transfer entropies at a significance level of 5%								
10	0.470	0.187	−0.557	0.079	0.499	0.054	−0.551	−0.261

(continued on next page)

Table C.3 (continued)

Size	(ALL) daily		(ALL) weekly		(EN) daily		(EN) weekly	
	Wiki Random		Wiki Random		Wiki Random		Wiki Random	
20	0.646	0.041	−0.629	−0.132	0.725	0.137	−0.422	−0.023
30	0.664	0.079	−0.668	0.039	0.849	−0.041	−0.383	0.186
40	0.689	0.110	−0.657	0.021	0.977	−0.092	−0.347	−0.283
50	0.774	0.000	−0.647	0.227	1.055	0.001	−0.323	−0.151

The table summarizes average cumulative portfolio returns (in basis points) of $k = 100$ Wiki- and random-portfolios with varying portfolio size m . Wiki- and random-portfolios are constructed based on the results of the Shannon transfer entropy estimation in Figs. 1(a) and 2(a), here denoted by (ALL) and (EN), respectively. Both a daily and weekly trading pattern is considered. The entire time span from 2008 to 2017 is used to calculate overall cumulative portfolio returns, which are then averaged across all k portfolios. Panel A refers to results of portfolios constructed based on the transfer entropy estimates depicted in Figs. 1(a) and 2(a). Panel B applies a more conservative significance level of 5% to the transfer entropy estimates on which portfolio construction is subsequently based.

Table C.4

Skewness of portfolio returns of Wiki- and random-portfolios in hypothetical trading strategy (2010–2017).

Size	(ALL) daily		(ALL) weekly		(EN) daily		(EN) weekly	
	Wiki Random		Wiki Random		Wiki Random		Wiki Random	
Panel A: transfer entropies at a significance level of 10%								
5	−0.120	−0.039	−0.541	0.084	−0.133	−0.019	−0.657	−0.047
10	−0.056	−0.011	−0.778	0.024	−0.124	−0.032	−0.891	−0.011
15	0.010	−0.004	−1.029	−0.177	−0.046	0.019	−1.023	−0.031
20	0.045	−0.029	−1.098	−0.019	0.005	0.021	−1.178	0.117
25	0.070	0.011	−1.166	−0.143	0.064	0.033	−1.255	0.078
Panel B: transfer entropies at a significance level of 5%								
5	−0.034	0.018	−0.701	−0.042	−0.026	−0.018	−0.674	0.107
10	0.046	−0.070	−0.913	0.110	0.024	0.048	−1.005	0.133
15	0.136	0.025	−1.062	0.164	0.120	−0.002	−1.111	0.138
20	0.192	0.051	−1.162	−0.091	0.176	0.013	−1.213	0.038
25	0.277	0.039	−1.243	0.020	0.226	0.020	−1.273	−0.066

The table summarizes average cumulative portfolio returns (in basis points) of $k = 100$ Wiki- and random-portfolios with varying portfolio size m . Wiki- and random-portfolios are constructed based on the results of the Shannon transfer entropy estimation in Figs. 1(b) and 2(b), here denoted by (ALL) and (EN), respectively. Both a daily and weekly trading pattern is considered. The time span from 2010 to 2017 is used to calculate overall cumulative portfolio returns, which are then averaged across all k portfolios. Panel A refers to results of portfolios constructed based on the transfer entropy estimates depicted in Figs. 1(b) and 2(b). Panel B applies a more conservative significance level of 5% to the transfer entropy estimates on which portfolio construction is subsequently based.

Table C.5

Kurtosis portfolio returns of Wiki- and random-portfolios in hypothetical trading strategy (2008–2017).

Size	(ALL) daily		(ALL) weekly		(EN) daily		(EN) weekly	
	Wiki Random		Wiki Random		Wiki Random		Wiki Random	
Panel A: transfer entropies at a significance level of 10%								
10	24.180	23.650	15.606	16.988	24.318	23.942	15.132	16.290
20	24.366	18.801	13.259	17.030	26.230	22.174	13.316	14.646
30	25.702	18.126	12.617	17.587	26.528	20.726	12.444	16.686
40	26.084	19.369	12.486	17.490	27.849	21.408	12.018	16.696
50	27.034	17.213	11.990	15.360	27.634	18.652	11.608	16.071
Panel B: transfer entropies at a significance level of 5%								
10	20.510	21.197	13.549	17.416	23.039	23.465	14.768	16.605
20	21.929	18.720	12.359	15.495	24.937	20.818	13.513	14.482
30	22.793	18.865	11.685	16.636	25.932	23.183	12.798	15.592
40	23.076	19.245	11.520	13.275	27.565	21.335	12.487	15.647
50	23.417	18.066	11.107	16.754	27.788	19.984	12.192	16.879

The table summarizes average cumulative portfolio returns (in basis points) of $k = 100$ Wiki- and random-portfolios with varying portfolio size m . Wiki- and random-portfolios are constructed based on the results of the Shannon transfer entropy estimation in Figs. 1(a) and 2(a), here denoted by (ALL) and (EN), respectively. Both a daily and weekly trading pattern is considered. The entire time span from 2008 to 2017 is used to calculate overall cumulative portfolio returns, which are then averaged across all k portfolios. Panel A refers to results of portfolios constructed based on the transfer entropy estimates depicted in Figs. 1(a) and 2(a). Panel B applies a more conservative significance level of 5% to the transfer entropy estimates on which portfolio construction is subsequently based.

Table C.6

Kurtosis of portfolio returns of Wiki- and random-portfolios in hypothetical trading strategy (2010–2017).

Size	(ALL) daily		(ALL) weekly		(EN) daily		(EN) weekly	
	Wiki	Random Wiki	Random Wiki		Random Wiki		Random	
Panel A: transfer entropies at a significance level of 10%								
5	8.674	8.504	11.843	9.034	9.008	9.189	9.926	8.412
10	9.819	7.916	13.162	9.403	9.012	8.007	11.448	7.562
15	10.774	7.975	14.901	8.115	9.664	7.116	12.453	8.077
20	11.707	7.461	15.363	8.704	10.033	7.058	13.057	7.797
25	12.533	7.354	15.640	8.330	10.807	6.977	13.708	8.255
Panel B: transfer entropies at a significance level of 5%								
5	8.818	8.510	12.782	8.843	8.211	8.626	10.582	9.094
10	9.895	7.737	14.514	8.055	9.210	7.529	12.735	10.149
15	11.182	7.914	15.631	9.955	9.590	7.021	12.869	9.428
20	11.927	8.102	15.789	8.782	10.438	7.129	13.192	8.839
25	12.842	7.581	16.097	8.123	10.888	7.163	13.412	9.586

The table summarizes average cumulative portfolio returns (in basis points) of $k = 100$ Wiki- and random-portfolios with varying portfolio size m . Wiki- and random-portfolios are constructed based on the results of the Shannon transfer entropy estimation in Figs. 1(b) and 2(b), here denoted by (ALL) and (EN), respectively. Both a daily and weekly trading pattern is considered. The time span from 2010 to 2017 is used to calculate overall cumulative portfolio returns, which are then averaged across all k portfolios. Panel A refers to results of portfolios constructed based on the transfer entropy estimates depicted in Figs. 1(b) and 2(b). Panel B applies a more conservative significance level of 5% to the transfer entropy estimates on which portfolio construction is subsequently based.

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